

RESEARCH ARTICLE

Machine Learning-Driven Discovery of Thermoset Shape Memory Polymers With High Glass Transition Temperature Using Variational Autoencoders

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ABSTRACT

The discovery of high-performance shape memory polymers (SMPs) with enhanced glass transition temperatures (T_g) is of paramount importance in fields such as geothermal energy, oil and gas, aerospace, and other high-temperature applications, where materials are required to exhibit shape memory effect at extremely high-temperature conditions. Here, we employ a novel machine learning framework that integrates transfer learning with variational autoencoders (VAE) to efficiently explore the chemical design space of SMPs and identify new candidates with high T_g values. We systematically investigate the effect of different latent space dimensions on the VAE model performance. Several machine learning models are then trained to predict T_g . We find that the SVM model demonstrates the highest predictive accuracy, with R^2 values exceeding 0.87 and a mean absolute percentage error as low as 6.43% on the test set. Through systematic molar ratio adjustments and VAE-based fingerprinting, we discover novel SMP candidates with T_g values between 190°C and 200°C, suitable for high-temperature applications. These findings underscore the effectiveness of combining VAEs and SVM for SMP discovery, offering a scalable and efficient method for identifying polymers with tailored thermal properties.

1 | Introduction

The range of materials available on Earth is enormous, with projections suggesting there could be close to a million unique types [1]. This number might even exceed initial estimates due to the discovery of novel crystalline phases in inorganic materials and the complex, varied structures seen in polymers [2]. The rapid identification of promising candidate structures for viable materials has been significantly enhanced by the use of machine learning (ML), which demonstrates notable accuracy and efficiency across a wide range of fields, including proteins [3–5], metals [6, 7], organic molecules [8, 9], and polymers [10–15].

Nevertheless, discovering new polymer materials is a complicated task, primarily due to the limited datasets. Since the synthesis and property evaluation of a single polymer can take over a month or even over a year, the quantity of available material data often remains confined to just a handful of samples.

ML has been extensively utilized across various materials, particularly in the pharmaceutical sector. Over 10 databases have been created specifically for drug research. PDBbind [16] offers data on receptor-ligand interactions for proteins with known structures, while Pubchem [17] consists of a broad spectrum of chemical data, including data on physicochemical properties

and biomolecular interactions. Uniprot [18] provides extensive data on protein sequence similarity and protein identification. RCSB PDB [19] offers insights into the three-dimensional structures of proteins. Collectively, these resources encompass data on tens of millions of molecules. The similarities between drugs and polymers, such as molecular complexity and the need for detailed structural information, make these drug databases particularly useful for polymer research as well. Researchers have leveraged these similarities to apply ML techniques developed for drugs to polymers, as demonstrated in studies by Bannigan et al. [20] and Yan et al. [13], who utilized drug datasets to develop polymer ML models. The application of ML frameworks in polymer discovery, particularly for shape memory polymers (SMPs) with high glass transition temperature (T_g), offers several key advantages over traditional experimental and computational methods. Traditional approaches to discovering novel SMPs often involve labor-intensive synthesis and testing, which are not only time-consuming but also limited in scope due to resource constraints. In contrast, ML accelerates this process by enabling rapid exploration of the vast chemical design space. Moreover, ML models excel at capturing complex, non-linear relationships between molecular features and material properties, delivering higher accuracy and reliability compared with conventional computational methods. This framework not only reduces the dependency on extensive datasets but also offers scalability, making it a powerful tool for navigating the intricate relationships between polymer composition and performance.

Variational autoencoders (VAEs) are a powerful ML tool increasingly applied in polymer discovery, enabling the generation and exploration of new molecular structures [21, 22]. Their ability to map complex molecular data to a lower-dimensional latent space facilitates the discovery of novel polymers with specific properties. The performance of a VAE is highly dependent on the quality and diversity of the dataset used for training. For instance, Batra et al. [23] trained a VAE using 250,000 Simplified Molecular Input Line Entry System (SMILES) to discover polymers for extreme conditions. Additionally, Lee et al. [24] leveraged 5473 alloy data points to train their VAE. For SMPs; however, the availability of large, well-structured datasets is often a limitation, which can hinder model performance. To overcome this challenge, a transfer learning-variational autoencoder (TL-VAE) ML model has been implemented. We utilized a large drug dataset to pre-train the VAE model, taking advantage of the molecular similarities between drugs and polymers. Following this, we fine-tuned the model using a manually curated SMPs dataset to adapt the latent space and improve the model's predictive accuracy for SMPs properties.

The relevance of SMPs extends beyond traditional applications, emerging as pivotal in sectors demanding adaptive materials with precise control over mechanical properties [25–30]. Specifically, polymers with a high T_g are instrumental in high-temperature applications such as in geothermal power generation and oil and gas drilling, in which the shape memory effect (SME) of SMPs can be leveraged as proppants and loss circulation materials [31–38], where SMPs must exhibit SME under the high downhole temperature and pressure [39].

SMPs stand out in this field for their exceptional ability to return to their original shape after deformation, a feature that can

greatly improve both efficiency and longevity in the geothermal and oil and gas industries. The T_g is a critical property in SMPs, particularly in amorphous SMPs where it governs the thermal-responsive behavior and mechanical adaptability. At temperatures below T_g , the polymer chains are in a glassy, rigid state, where molecular mobility is limited. As the temperature approaches and exceeds T_g , the polymer transitions into a rubbery state, characterized by increased chain mobility and segmental motion, which facilitates the SME. This transition is crucial for SMPs, as it determines their ability to recover their original shape after deformation [40, 41]. Moreover, SMPs with high T_g are crucial in the aerospace industries due to their ability to withstand extreme thermal conditions while maintaining their shape-memory properties [42]. The ability to tailor the T_g and mechanical properties of SMPs enables the development of more efficient, adaptive aerospace systems, improving fuel economy, and reducing the weight of aircraft, satellites, and space vehicles [43–45]. Therefore, designing new SMPs with high T_g is highly desired. However, the discovery and optimization of high T_g SMPs involves navigating complex and large compositional space, making ML an invaluable tool in identifying and predicting the properties of these advanced materials efficiently.

While ML offers transformative potential in the discovery and optimization of SMPs, the application of these techniques faces significant challenges, primarily due to the scarcity of comprehensive datasets. The development of high T_g SMPs is a complex process, involving nuanced interplays of polymer chemistry and physical behaviors under varying environmental conditions. Recently, in-depth studies have been done on exploring SMPs with optimum properties. For instance, Yan et al. [46] developed a dual-convolutional neural network (CNN) to predict the T_g and recovery stress of SMPs through forward design. Additionally, in our previous investigations, the weighted-vector combination method (WVCM) was applied in conjunction with a VAE model to produce novel SMPs with enhanced performance (controlling T_g and rubbery modulus E_r) [13]. Moving forward, it is imperative to enhance the accuracy of this model by addressing the limitations.

The previous model has been trained on 420,000 drug samples and then fine-tuned by 245 thermoset SMPs [13]. Moreover, the encoder of the VAE model consists of CNN layers and dense layers. On the other hand, the decoder is composed of a long short-term memory neural network instead of CNN layers to capture the sequence in the molecular structures. To enhance the model's generative power and robustness, expanding the dataset and implementing targeted refinements in the VAE model are crucial. Furthermore, one of the key parameters influencing the accuracy of VAEs is the dimensionality of the latent space, as it directly affects the ability of the model to capture and represent complex features in the data [47]. A well-chosen latent space dimension balances the model's ability to generalize without overfitting, allowing for the efficient encoding of meaningful patterns while maintaining computational efficiency. Optimizing this parameter is essential to improve reconstruction quality and predictive performance [48]. Furthermore, data augmentation is essential in the discovery of new polymers, especially in minimizing dependence on training data and improving the performance of ML models [49]. By leveraging data augmentation techniques,

including generating structurally similar SMILES representations using tools like RDKit, we can effectively expand the available dataset [50]. For instance, data augmentation techniques have been used to improve the accuracy of ML models to identify environmental polymers and microplastics [51]. In tandem with optimizing the latent space and refining VAE architecture, this approach creates a comprehensive strategy for advancing the model's performance.

In this investigation, we employed a combination of a VAE model and WVCN to discover SMPs with high Tg. The WVCN method has been utilized to capture the impact of molar ratio variations between the monomer and the cross-linker. In our framework, the fingerprints of the molecular structures are generated using the VAE model, allowing for efficient exploration of the latent space to discover SMPs with desirable Tg. The novelty of this work stems from several key advancements. First, the dataset for model training was significantly expanded, utilizing a larger drug dataset for training and an increased number of SMPs for fine-tuning. Second, we introduced substantial modifications to the VAE architecture, particularly focusing on hyperparameter tuning and an extensive refinement of the encoder and decoder, which now feature more sophisticated mechanisms for generating SMPs. Third, the effect of molar ratio of each monomer on the SMP's thermal property has been captured by the ML model and validated experimentally. Furthermore, we conducted a detailed study on the effect of latent space dimensionality on the model's predictive accuracy, identifying an optimal configuration that balances computational efficiency with high reconstruction quality. Finally, one of the potential SMPs with high Tg has been selected for the experimental test to validate the accuracy of the ML framework. These innovations collectively enhance the generative and predictive capabilities of the VAE, making this framework versatile and suitable for discovering and optimizing SMPs with not only high Tg but also other key properties, such as mechanical strength, recovery stress, and thermal stability.

2 | Method

Here, we introduce the three main parts of the SMP discovery framework with high Tg. In this study, VAEs were utilized to fingerprint SMILES representations of monomers and polymers. The VAE model excels in capturing and encoding high-dimensional data into a compact latent space, enabling efficient representation of molecular features. This capability is particularly advantageous for uncovering patterns and relationships within complex chemical datasets, which are critical for accurately predicting Tg. These parts include the development of the TL-VAE model, the establishment of SMP Tg prediction models, and the development of the SMP forward design model. This framework with its constituents is presented in Figure 1.

The specific procedures are outlined as follows:

A. Development of the TL-VAE model.

1. A dataset of 550,000 drug molecules is used to train the VAE model (a), which is about 130,000 more data points compared with the previous model (420,000 drug molecules) [13], to obtain the intermediate latent space.

2. To optimize the accuracy of the VAE model, seven various VAE models with different latent sizes are trained.
3. The VAE model is fine-tuned (b) using 289 SMPs to establish the final latent space.

B. Development of the SMPs Tg predictor models.

1. Each monomer in an SMP network is fingerprinted into its high-dimensional vector in the latent space using the TL-VAE model (e).
2. The high-dimensional vector of each monomer is multiplied by the corresponding molar ratio (f). The summation of the weighted vectors of the monomers produces a high-dimensional vector for each SMP.
3. Utilizing the high-dimensional vectors as inputs with the Tg of each SMP as the corresponding output, three ML models including CNN, SVM, and RF, are trained to predict the Tg of the SMPs (c).

C. Development of the SMP forward design model.

1. Combining the monomers and the crosslinkers randomly with different molar ratios (ranging from 5% to 95% in 5% intervals) to generate new SMPs (h).
2. Generating a high-dimensional vector for each SMP using the TL-VAE model (i).
3. The best ML model is selected to predict the Tg of the generated SMPs (j).
4. The SMPs that their Tg are in the desired range are selected for experimental test to validate the model, and the desired SMP is discovered (k).

2.1 | Developing an Optimum VAE Model

VAEs, first introduced by Kingma and Welling in 2013 [52], are a generative model that leverages deep learning to efficiently encode data into a continuous latent space while maintaining stochasticity. Unlike traditional autoencoders, VAEs introduce a probabilistic approach to modeling, allowing for the generation of new data points by sampling from the latent space. This innovation has significantly influenced various fields, including polymer discovery and molecular design, due to its capacity to learn meaningful data representations. For instance, Abeer et al. [53] used VAEs to develop a molecular design framework. They used a multi-objective latent space optimization technique to improve the sampling efficiency of the model. Furthermore, Kim et al. [54] leveraged VAEs to develop a generative model to discover optimal polymers which show the constituent reactants, and reaction pathways.

To facilitate ML on chemical structures, we employed a systematic approach for encoding molecular information. In Figure 1a, chemical structures were initially encoded using canonical SMILES notation. These SMILES strings were subsequently converted into binary matrices of dimensions 204×55 through a one-hot encoding method. In this context, 204 denotes the longest SMILES string length, while 55 signifies the number of characters in the encoding set (as detailed in Figure 2 of reference [46]). The resulting binary matrices are visualized in the form of grayscale images, which makes it logical to apply a CNN for analysis—a technique widely used for learning images. Afterward, the matrices were passed through the VAE model to generate the latent space. The encoder is responsible

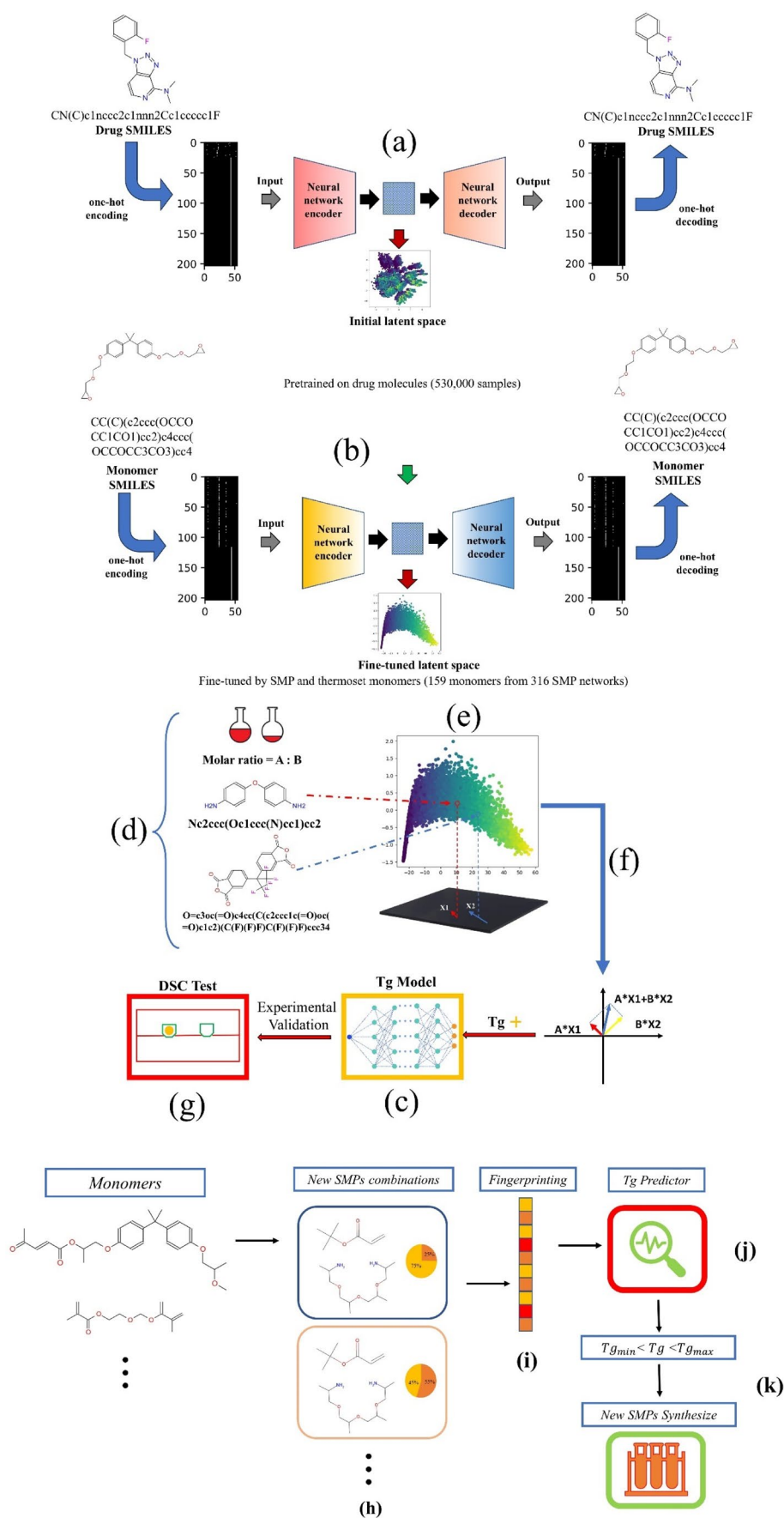


FIGURE 1 | Schematic representation of the framework used for discovering SMPs with user-defined Tg.

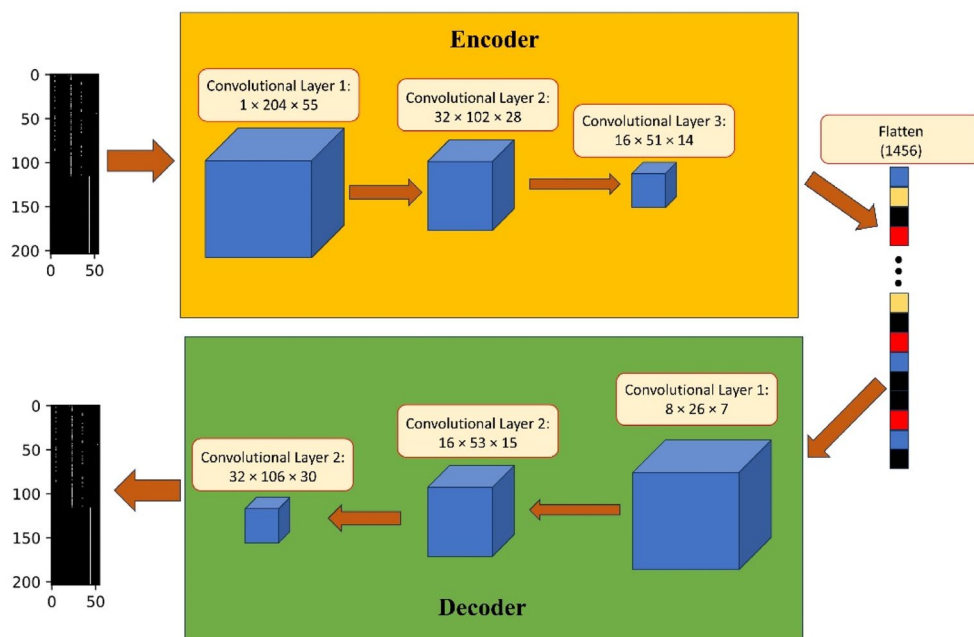


FIGURE 2 | Illustration of the encoder–decoder architecture of the VAE model.

for converting the matrices into latent vectors of a specified dimension, while the decoder regenerates these vectors into the original 204×55 matrix structure.

VAEs can be constructed using various neural network architectures, each tailored to specific tasks. For example, some VAEs employ CNNs in the decoder to improve the reconstruction of spatially structured data [13], while others use Recurrent Neural Networks (RNNs) to handle structured data [55], particularly SMILES strings or time-series inputs. RNN-based VAEs are particularly effective when dealing with sequences due to their ability to capture temporal dependencies. Additionally, Deep Neural Networks (DNNs) are sometimes incorporated into the encoder to extract abstract, high-level features from complex input data [56]. Each architecture brings unique advantages, such as CNNs excelling at local feature extraction and RNNs efficiently modeling sequential patterns. In our study, we opted for a model that uses CNNs in both the encoder and decoder. This configuration capitalizes on the spatial nature of the SMILES-encoded binary matrices, allowing for more accurate feature extraction and reconstruction, thereby enhancing the overall performance of the model.

The dataset used in this study was meticulously gathered manually from published research papers, ensuring a comprehensive and diverse representation of SMPs. This approach allowed us to include a wide range of monomer compositions, molar ratios, and corresponding Tg, which are critical for training and validating the ML models. The dataset was divided into training and testing subsets, with 85% of the data allocated for training and 15% reserved for testing. This split was chosen to ensure that the models were provided with sufficient data for learning while maintaining an independent test set for evaluating their generalization performance. During preprocessing, SMILES representations were verified for consistency to avoid discrepancies in molecular structures. Molar ratios were carefully extracted to reflect the proportions of monomers in each polymer

TABLE 1 | Original and augmented SMILES representations of the same chemical structure.

SMILES	Version
C(COCCOCCOCCOC(=O)C(=C)C) OCCOCCOCCOCCOC(C(=C)C)=O	Original
C(COCCOCCOCCOCCOCCOCCOC(=O) C(C)=C)OCCOC(=O)C(=C)C	Augmented
C(COCCOCCOCCOCCOCCOCCOCCOC (=O)C(=C)C)OC(=O)C(=C)C	Augmented
O(CCOCOCOCOCOCOC(=O)C(C)=C) CCOCCOCCOC(=O)C(=C)C	Augmented

system. The chemical structures of SMPs are quite intricate, and yet we were able to gather only 159 unique monomers from a total of 300 SMPs. This limited dataset makes it challenging for the VAE to effectively learn the SMILES syntax, let alone the specific atomic arrangements of the SMP monomers. To address this challenge, we adopted a two-phase training approach. Initially, we compiled a large dataset of 550,000 small molecules from ChEMBL, an open-source biological database of drug-like molecules. With this extensive dataset, the VAE was able to grasp the general SMILES syntax, the basic structures of small molecules, and the specific atomic arrangements of compounds that share similarities with thermoset shape memory polymer (TSMP) monomers. In the second phase, we fine-tuned the model to specialize in TSMP monomers. Given the small dataset of only 159 unique monomers, we employed data augmentation techniques. Using RDKit, an open-source cheminformatics tool, we generated possible SMILES representations for each concatenated monomer, significantly expanding the dataset to 158,114 distinct SMILES. It is worth noting that the various SMILES strings representing the same monomer structure only exhibit minor variations. For instance, Table 1 indicates a unique

monomer with three varying SMILES representations. In the second phase of training, we fine-tuned the model by updating only the weights and biases of the decoder, while keeping the rest of the model frozen.

Optimizing the VAE model involved investigating the impact of different latent space sizes, which play a critical role in the model's performance. The latent space serves as a compressed representation of the molecular data, and its dimensionality directly affects the balance between capturing essential features and avoiding overfitting. A too-small latent space might not encode enough information to reconstruct the input accurately, while a too-large latent space could lead to overfitting and reduce the model's generalizability [57]. In our study, we experimented with various latent space dimensions to find the optimal balance, ensuring that the VAE could effectively capture the intricate patterns in the SMP monomers while maintaining the ability to generalize to new structures. A cluster (SuperMIC) consisting of two NVIDIA V100 16GB GPU has been used to train our models.

In order to illustrate the accuracy of our training strategy, we utilized the VAE model and principal component analysis (PCA) to transform the chemical structures of the TSMP monomers into 2D vectors. This high-dimensional mapping is crucial, as the subsequent Tg predictor model relies on the output from the VAE. We aim for the VAE model to capture key features in the latent space, which are expected to enhance the performance of the Tg predictor in predicting outcomes. Figure 3 compares the PCA mapping of the trained VAE models with various latent sizes. The latent space encodes the high-dimensional input data into a smaller, more abstract representation, which is crucial for fingerprinting the chemical structures. The latent size represents the number of features that the model compresses the data into. In Figure 3a, the latent space is relatively condensed

(latent size = 16). The limited latent size forces the VAE model to distill the input into a much smaller representation, which can lead to a bottleneck. As a result, the variation between data points is not well captured, leading to tighter clusters and fewer distinct regions in the latent space. On the other hand, The PCA plot for the model with latent size 1024 (Figure 3b) shows a much more spread-out distribution. This indicates that the model is capturing significantly more variation and nuance between data points. With a larger latent size, the model has more capacity to encode detailed information about each input. However, this comes with the risk of overfitting, as the model might over-represent specific data points. The PCA mappings for the VAE models with latent sizes between 16 and 1024 are presented in Figure S1.

The VAE model was trained using a custom training loop that optimizes the reconstruction loss and Kullback–Leibler (KL) divergence. The model encodes input SMILES strings into latent variables and reconstructs them, with the reconstruction loss calculated using categorical cross-entropy between the original and decoded SMILES. The KL divergence is computed to regularize the latent space by penalizing deviations from a Gaussian distribution. A validity check is also performed for each decoded SMILES string, applying a penalty based on how many valid SMILES were generated, which is incorporated into the final loss. The validity check is an essential component of the training process for ensuring that the generated SMILES strings represent valid chemical structures. During each training batch, decoded SMILES strings are evaluated by converting the predicted tensor into a SMILES format and using the RDKit library to check whether the SMILES corresponds to a valid molecule.

The advantage of incorporating this validity check is that it directly encourages the model to generate chemically meaningful structures. Without this check, the VAE could focus solely on

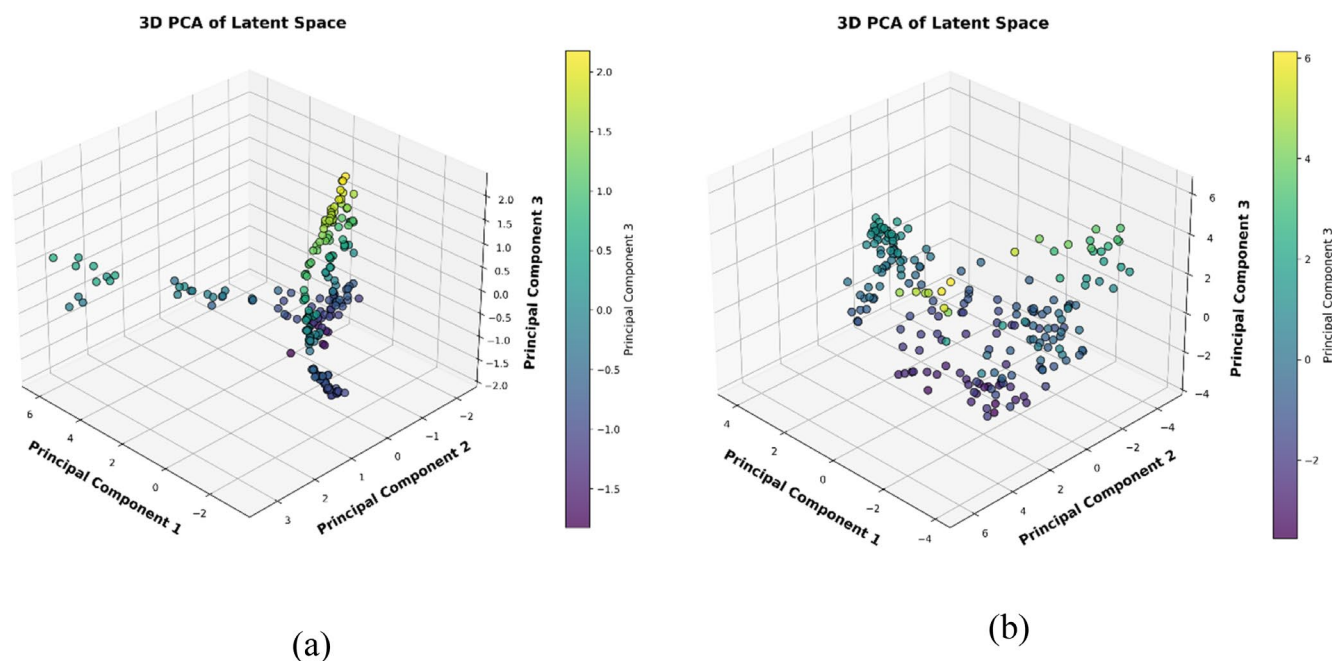


FIGURE 3 | Two PCA mapping of VAE models by using various latent sizes (a) 16 and (b) 1024.

minimizing the reconstruction (Equation 1) loss, potentially generating SMILES strings that are syntactically correct but chemically nonsensical. The loss functions employed in the VAE models are expressed as follows:

$$L(\hat{y}, y) = -\frac{1}{N} \sum_i^N [y_i \log \hat{y}_i + (1 - y_i) \log(1 - \hat{y}_i)] \quad (1)$$

In this context, y_i represents the predicted outcome, while $\hat{y}_i$ denotes the actual ground truth value. N corresponds to the total number of training data points. Moreover, μ_i and σ_i are the mean and standard deviation for the latent variables, and d is the dimensionality of the latent space. Table 2 presents the hyperparameters in the VAE model and the VAE schematic is illustrated in Figure 2. In this study, the selection of parameters for the VAE model was conducted manually through an iterative process. Key hyperparameters, such as the number of layers, latent size, learning rate, and batch size, were adjusted based on their observed

effects on model performance, as evaluated by the reconstruction loss and prediction accuracy for Tg.

Cosine similarity calculates the angle between two vectors in a multi-dimensional space to assess their similarity, independent of their size or magnitude. A higher cosine similarity between vectors indicates that the VAE has successfully captured and encoded essential features, resulting in a more meaningful latent representation. The average cosine similarity can be calculated using Equation (2):

$$\text{Average cosine similarity} = \frac{\sum_{i=1}^n \frac{V_i \cdot V_o}{\|V_i\| \|V_o\|}}{N} \quad (2)$$

where in this equation V_i and V_o are indicating the input and output matrices, respectively. To show the effect of latent size on the average cosine similarity of the VAE models, Table 3 is presented. Figures S2 and S3 illustrate the input binary matrices and their corresponding decoded matrices of the pre-trained and fine-tuned models, respectively. Additionally, the algorithms for developing the pre-trained and fine-tuned VEA models are presented in the Supporting Information: Algorithms 1 and 2.

The results presented in Table 3 demonstrate a clear relationship between the latent size and the average cosine similarity of the VAE models. The VAE models are trained using the drug molecules dataset and then fine-tuned using the expanded SMPs dataset. While the similarity in the pre-trained models is close to their corresponding fine-tuned models, we can see a slight increase in the similarity. First, since this similarity check has been done using the SMPs dataset, the results show the resemblance of drug molecules to SMPs. It demonstrates the effectiveness of our strategy. Second, the average similarity of the fine-tuned models increased with respect to their corresponding pre-trained models. This indicates that the transfer learning strategy has been successful in improving the VAE model's ability to capture more details of the SMPs.

Furthermore, as the latent size increases, the average cosine similarity between the input and output SMILES vectors (V_i and V_o) improves significantly. This indicates that larger latent sizes allow the VAE model to capture more intricate details and features of the input data, leading to more accurate reconstructions. For instance, the model with a latent size of 16 achieves

TABLE 2 | Architecture details and hyperparameters of the VAE model.

Hyperparameters	Values or items
Batch size	512
Latent sizes	16, 32, 64, 128, 256, 512, 1024
Learning rate	0.0001
Optimizer	Adam
Encoder layers	Two-dimensional convolutional layer: 1-32-16-8, Fully connected: 1456
Decoder layers	Fully connected: 1456, Two-dimensional convolutional layer: 8-16-2-1
Activation function	ReLU
Kernel size	3 × 3

TABLE 3 | The average cosine similarity of VAE models with various latent sizes.

Latent size (pre-trained models)	Average cosine similarity (%)	Latent size (fine-tuned models)	Average cosine similarity (%)
16	82.06	16	82.53
32	83.23	32	83.96
64	86.49	64	88.78
128	89.46	128	92.50
256	95.61	256	97.14
512	97.76	512	99.09
1024	99.20	1024	99.71

an average cosine similarity of 82.53%, indicating that while it captures the general structure, some finer details are lost. In contrast, with a latent size of 1024, the model achieves an impressive 99.71% similarity, demonstrating that almost all features of the input are preserved in the output.

2.2 | SMP Tg Predictor

The WVCM [13] was used to make fingerprints for the SMPs. Using this method, we were able to use our datasets to train various models for SMP Tg prediction. We divided the dataset into training (85%) and testing (15%) subsets to ensure unbiased model evaluation. We used three ML models to predict the Tg of the polymers. The models were CNN, support vector regression (SVR), and random forest (RF). These models were selected for their complementary strengths in capturing both linear and non-linear relationships within the data, handling high-dimensional feature spaces, and extracting intricate patterns from the input representations. CNNs are particularly adept at recognizing spatial hierarchies and local dependencies in data, making them well-suited for processing molecular fingerprints. SVR, with its robust ability to model complex relationships using kernels, is ideal for regression tasks involving non-linear dependencies. RF, a powerful ensemble learning method, excels in managing variability in the dataset and preventing overfitting. Moreover, the algorithm for predicting the Tg of SMPs using various ML models is presented in the [Supporting Information](#): Algorithm 3.

2.2.1 | CNN Model

We utilized a CNN model due to its powerful capability to automatically extract and learn hierarchical features from complex input data. CNNs are particularly well-suited for identifying patterns and spatial hierarchies in data, making them ideal for handling the intricate structural information present in polymer molecules. Table S1 shows the hyperparameters of the trained CNN model. The mean squared error (MSE) was used as the loss function which can be written as:

$$\text{MSE} = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2 \quad (3)$$

In this equation, $\hat{y}_i$ represents the model's predicted value, while y_i corresponds to the actual target value for each instance. The term N refers to the total number of training samples in the dataset.

2.2.2 | SVR Model

We employed a support vector machine with a regression approach (SVR) for predicting the (Tg) of SMPs. The SVR model, using a radial basis function (RBF) kernel, was selected due to its robust ability to handle non-linear relationships between features and target variables, which are crucial when modeling complex molecular structures. The hyperparameters for the SVR model are presented in Table S2. SVR tries to find a function $f(x)$

that has at most ϵ -deviation from the actual target values y for all training data while maintaining model simplicity. The SVR function can be expressed as:

$$f(x) = \sum_{i=1}^n (\alpha_i - \alpha_i^*) K(x_i, x) + b \quad (4)$$

where x represents the input features, α_i and α_i^* are the Lagrange multipliers (support vectors) determined by solving the optimization problem, $K(x_i, x)$ is the kernel function, and b is the bias term.

2.2.3 | RF Model

RF regressor was utilized due to its robustness in handling non-linear relationships and reducing overfitting by averaging multiple decision trees. RF are particularly advantageous because they aggregate predictions from several independent trees, leading to improved generalization and stability in the model's performance. Each tree in the forest is trained on a different subset of the data, with the final prediction being the average of the individual predictions. The key hyperparameters used for this model include 100 decision trees. The final prediction of the RF regression is the average of the predictions from each individual decision tree:

$$\hat{y} = \frac{1}{T} \sum_{t=1}^T h_t(x) \quad (5)$$

In this equation, T is the total number of decision trees, $h_t(x)$ is the prediction of the t -th decision tree for the input features, and $\hat{y}$ is the predicted output.

To showcase the effectiveness of our approach and compare our results with previous investigations, we established a benchmark for comparative analysis. We used the VAE models to generate fingerprints for the monomers and then, leveraging the MVCM method, each SMP has a unique vector. These vectors are used to develop ML models to predict the Tg of the SMPs. Table 4 highlights the coefficient of determination (R^2) and mean absolute percentage error (MAPE) of the VAE combined with SVR models, which were chosen for presentation due to their superior performance compared to the other models. The accuracy of the other models is detailed in the [Supporting Information](#). The R^2 and MAPE are calculated using the following equations:

$$R^2 = 1 - \left(\frac{\sum_{i=0}^N (y_i - \hat{y}_i)^2}{\sum_{i=0}^N (y_i - \bar{y})^2} \right) \quad (6)$$

$$\text{MAPE} = \frac{1}{N} \sum_{i=1}^N \frac{|y_i - \hat{y}_i|}{y_i} \quad (7)$$

In addition, another evaluation metric called “percentage of correct point” (PCP) has been used to compare the results with our previous findings. The term “correct” refers to instances where the prediction falls within an acceptable range

TABLE 4 | The accuracy of ML models combined with SVR and different VAEs.

ML method	PCP training dataset (%)	PCP test dataset (%)	MAPE training dataset (%)	MAPE test dataset (%)	R^2 training	R^2 test
VAE (16)+SVR	88.01	69.77	5.88	10.41	0.97	0.70
VAE (32)+SVR	91.73	74.42	4.85	9.73	0.98	0.76
VAE (64)+SVR	93.39	83.72	3.97	6.43	0.98	0.87
VAE (128)+SVR	93.80	88.37	3.55	6.63	0.98	0.84
VAE (256)+SVR	95.45	86.04	3.07	7.03	0.98	0.86
VAE (512)+SVR	96.28	83.72	2.76	7.41	0.99	0.88
VAE (1024)+SVR	98.35	74.41	1.73	10.76	0.99	0.86
Old Model (VAE [256] + CNN)	91.33	71.43	7.38	13.91	—	—

of difference from the actual ground truth value. The acceptable error in this study is selected to be 15% to have a better comparative investigation. In addition, we present the outcomes of our earlier model for comparative analysis. The VAE (64)+SVM model stands out as the best-performing model in terms of Test MAPE, achieving the lowest error (6.43%). This indicates that using a latent size of 64 in combination with SVM offers the most accurate predictions, making it a strong candidate for further investigation and potential use in the downstream task. The latent size plays an important role in the accuracy of the VAE+SVR model (in a similar way for other models), as reflected by the changes in MAPE across different latent sizes. In our results, as the latent size increases initially, the MAPE decreases, reaches a minimum, and then starts to increase again.

For smaller latent sizes (e.g., 16 and 32), the model exhibits higher MAPE values (10.41% and 9.73%, respectively). This indicates that the model is unable to capture the full complexity of the data with such a limited number of latent dimensions. Essentially, the model lacks the capacity to represent the important features necessary for accurate predictions. As the latent size increases to 64, the MAPE drops to its lowest value of 6.43%, suggesting that this latent size strikes an optimal balance between complexity and generalization. At this point, the model has enough capacity to capture the key underlying patterns in the data without overfitting, resulting in improved prediction accuracy. However, when the latent size is increased beyond 64, the MAPE begins to rise. This rise in error indicates that the model is beginning to overfit as it gains too much capacity to represent the training data. With more dimensions, the model starts to capture noise in the training set, leading to poorer generalization of the test data.

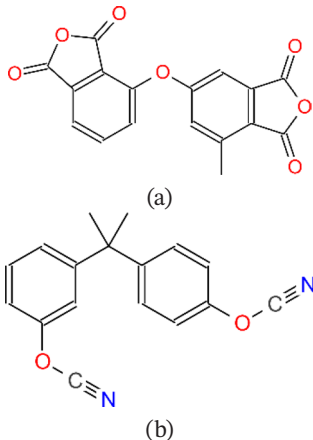
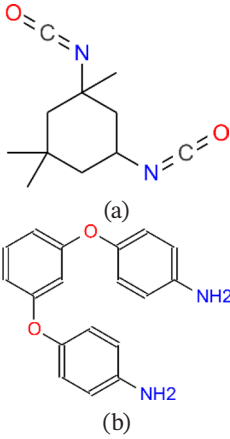
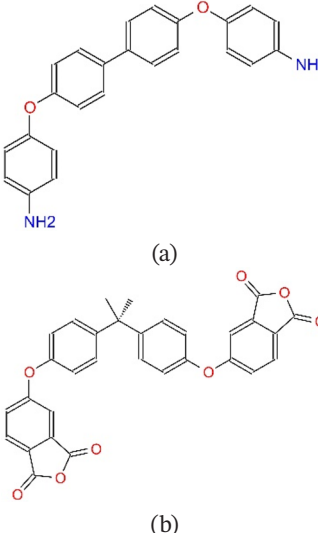
Moreover, the comparison between our best model, VAE (64)+SVM, and the old model reveals substantial improvements in predictive performance. The VAE (64)+SVM model achieved a significantly lower MAPE of 6.43%, compared to 13.91% for the old model, indicating more accurate predictions. Additionally, the test PCP improved from 71.43% to 83.72%, demonstrating better

generalization to unseen data. These results clearly show that the VAE (64)+SVM model is more reliable and effective in predicting the Tg of polymers than the previous model, making it a superior choice for this task. An additional advantage of the new model is its lower latent size compared to the old model (the latent size in the old model = 256). Despite using a smaller latent dimension (64), the model achieves superior performance with lower error, highlighting its efficiency in extracting relevant features. A lower latent size not only reduces the computational resources required, such as memory and processing time, but also accelerates both training and inference stages. This makes the model more practical for large-scale applications.

3 | Discovery of Novel SMPs

Using the selected model (VAE [64]+SVR), we discovered new SMPs based on the method outlined in Figure 1. Additionally, the algorithm for generating new SMPs with user-defined Tg is presented in the [Supporting Information](#): Algorithm 4. Our SMPs dataset comprises 159 unique monomers, each containing various functional groups. Monomers were systematically paired based on their functional groups, ensuring only chemically compatible combinations. This approach ensured the formation of feasible polymer structures. To generate new SMP candidates, we created high-dimensional vectors for the monomers using the VAE model. For each SMP, two monomers were selected, and their molar ratios were systematically varied from 5% to 95% in 5% increments. The high-dimensional vector of the first monomer was multiplied by its corresponding molar ratio, while the vector of the second monomer was multiplied by the remaining ratio, ensuring that their combined proportions always totaled 100%. Finally, the high-dimensional vectors of the two monomers, adjusted by their respective molar ratios, were summed to create a composite vector representing each SMP. Using the SVR model, we were able to predict the Tg of the new SMPs. Since we were focusing on the application of SMPs in geothermal power generation, the SMPs whose Tgs were in the range of 190°C to 200°C were selected. Table 5 presents the constituent monomers of three discovered SMPs and the corresponding molar ratios. The SMPs

TABLE 5 | Overview of discovered SMPs with monomer compositions, chemical structures, and Tg.

No.	SMILES/ratios	Chemical structure	Tg (°C)
1	<chem>['Cc3cc(Oc1cccc2c(=O)oc(=O)c12)cc4c(=O)oc(=O)c34', 'CC(C)(c1ccc(OC#N)cc1)c2cccc(OC#N)c2']</chem> (0.25, 0.75)	 (a) (b)	199.25
2	<chem>['CC1(C)CC(N=C=O)CC(C)(N=C=O)C1', 'Nc3ccc(OC2cccc(OC1ccc(N)cc1)c2)cc3']</chem> (0.4, 0.6)	 (a) (b)	196.05
3	<chem>[Nc4ccc(OC3ccc(c2ccc(OC1ccc(N)cc1)cc2)cc3)cc4', C[C@](C)(c3ccc(OC2ccc1c(=O)oc(=O)c1c2)cc3)c6ccc(OC5ccc4c(=O)oc(=O)c4c5)cc6']</chem> (0.45, 0.55)	 (a) (b)	192.24

No. 3 in Table 5 was synthesized to validate our model. The selection of monomers for this study was based on their commercial availability and the simplicity of their synthesis processes. These criteria were chosen to ensure that the materials could be readily obtained and easily synthesized, facilitating both experimental reproducibility and scalability. Additionally, the selected monomers exhibit functional groups suitable for achieving the desired polymer properties, aligning with the study's objectives.

The synthesis procedure, following the method by Kong et al. [58], involved placing 4.5 mmol of 4,4'-(1,1'-biphenyl-4,4'-diyl)dianiline (BPAB) in a three-neck flask with 40 mL of dimethylformamide. The solution was stirred under a nitrogen atmosphere until it completely dissolved. Afterward, 5.5 mmol of bisphenol A dianhydride (BPADA) was introduced, and the mixture was stirred continuously under nitrogen for 24 h at an ambient temperature to form poly(amic acid) (PAA). The PAA

TABLE 6 | Tg of synthesized SMPs at different BPAB/BPADA molar ratios.

Sample code	BPAB/BPADA ratios	Tg (experiment)		Tg (ML model)	Error
A	45%/55%	178.14°C		192.24°C	7.9%
B	50%/50%	222.07°C	229.6°C [58]	201.0°C	9.5%

was degassed in a vacuum chamber, poured onto a glass plate, and subjected to a stepwise thermal curing process, increasing the temperature at intervals of 70°C, 100°C, 150°C, 180°C, 220°C, and finally 260°C, with each stage held for 2 h. Next, in order to determine the Tg of the discovered SMP, differential scanning calorimetry (DSC) test was performed. The Tg was analyzed using a PerkinElmer 4000 DSC (MA, USA). Specimens weighing 5–10 mg were subjected to a heating and cooling cycle between 100°C and 240°C at a rate of 10°C/min, with a holding period of 2 min at both temperature extremes. The Tg value was determined from the second heating-cooling cycle. Throughout the procedure, nitrogen gas was purged at a consistent rate of 30 mL per minute. The calculated Tg of the SMP is presented in Table 6 and the heat flow versus temperature plots are illustrated in Figure S4.

Table 6 shows the Tg of the synthesized SMPs. While in the experimental test the Tg of sample A is 178.14°C, the prediction of the ML model is 192.24°C (7.9% error). The error is higher than the MAPE of the model on test dataset (6.43%) due to the lower number of data points at temperatures above 150°C. This makes it intricate to predict Tg above 200°C with high accuracy. To validate our experimental process, the sample with the same molar ratio (50%/50%) in the study done by Kong et al. [58] was synthesized. Sample B in Table 6 compares the Tg of our sample with that of the reference sample. It is important to note that Figure S4B shows the DSC heat flow versus temperature plot of our sample, while the reference study reported results using dynamic mechanical analysis (DMA). The observed 3.27% error may be attributed to potential experimental inaccuracies or inherent differences between DSC and DMA measurement techniques. DSC often identifies the onset of molecular mobility at the Tg but might not capture the full range of mechanical relaxation, leading to a lower Tg compared to DMA. Despite these discrepancies, the consistency of our sample's Tg with the reference study supports the validity of our experimental process.

The effect of the BPAB/BPADA molar ratio on the Tg is clearly demonstrated in both the experimental and ML-predicted results. As the molar ratio of BPAB increases (from sample A to B), the Tg rises correspondingly, indicating a strong correlation between the molecular composition and thermal properties of the synthesized SMPs. This rise in Tg can be attributed to the change in molecular interactions as the monomer composition alters, particularly a potential increase in strong π - π interactions and chain entanglements, which play a significant role in maintaining higher Tg values. This trend is well captured by the ML model, which accurately reflects the increasing Tg as the molar ratio shifts from 45%/55% to 50%/50%. The model's ability to capture this effect suggests that it successfully learns the underlying relationships between the molar ratio and the

resulting Tg. This implies that the model is proficient at understanding how molecular interactions—driven by varying monomer ratios—influence the macroscopic thermal behavior of SMPs.

4 | Conclusions

The integration of VAE with transfer learning has proven to be an effective strategy for the discovery of SMPs with high Tg values. By leveraging large drug datasets and fine-tuning the model with SMP data, we enhanced the model's ability to explore the chemical design space efficiently. The results demonstrate that optimizing the latent space significantly improves predictive accuracy, leading to the identification of new SMP candidates suitable for demanding applications such as geothermal energy and aerospace. These findings underscore the potential of our approach in addressing the challenge of limited data through effective transfer learning, and VAE optimization, which enhance the model's predictive capabilities. By optimizing the latent size, our model efficiently captures essential features without overfitting, ensuring accurate and computationally efficient predictions. Additionally, the incorporation of a chemical validity term into the VAE's loss function played a critical role in ensuring that the generated molecular structures are chemically meaningful. Furthermore, the proposed framework successfully accounts for the effect of molar ratios on Tg, demonstrating its robustness in handling complex chemical compositions and their influence on material properties. This comprehensive framework provides a scalable pathway for designing SMPs with tailored thermal properties and other critical characteristics. While this study highlights advancements in latent size optimization, chemical validity incorporation, and modeling the effect of molar ratios on Tg, it is limited by the dataset size and focus solely on Tg (other parameters such as recovery stress, thermal stability, or mechanical strength should be considered in future investigations). Additionally, the model assumes that molar ratios sufficiently capture interactions influencing Tg, leaving other factors for future exploration.

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Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.